Angu Praveen MS

CAE / Computational Engineer | FEA | CFD | Simulation | Automation | AI

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Visa: Permanent Resident | LinkedIn | Portfolio

Summary

CAE / Computational Engineer with 4+ years of experience delivering thermal, structural, and multiphysics simulation solutions across the semiconductor, automotive, and aerospace industries. Specialized in FEA, CFD, and automation using ANSYS, Abaqus, MATLAB, and Python. Proven impact includes reducing analysis time by **90%**, improving solver convergence by **20%**, and increasing thermal efficiency by **16%**.

Technical Skills

CAE / Simulation / Analysis:
FEA | CFD | Thermal Analysis | Structural Analysis | Static Structural | Heat Transfer | Fatigue Analysis | Design Validation

Simulation & Engineering Tools:
ANSYS Mechanical | ANSYS Fluent | ANSYS APDL | ANSYS Workbench | Abaqus CAE | Abaqus Scripting | OpenFOAM | SU2 | SUAVE | OpenVSP | ICEM CFD

Programming / Automation:
Python | MATLAB | Simulink | VBA | CAE Automation | APDL | SQL | C | C++

AI / ML & Advanced Concepts:
Explainable AI | Deep Learning | Physics-Informed ML | scikit-learn | Digital Twin | Condition Monitoring | Predictive Maintenance | ImageJ

Software & Environments:
Git | GitHub | SVN | Linux | macOS | Windows

Professional Experience

Independent Research Consultant – Predictive Maintenance & Digital Twin Sep 2025 – Present
External Collaboration with Amsterdam University of Applied Sciences, Amsterdam

- Developing an explainable Predictive Maintenance prototype and Digital Twin frameworks using legacy shipboard data to improve maintenance planning for marine auxiliary systems.
- Building interpretable AI models (GAM, EBM) for ship condition monitoring applications.
- Developing operator dashboards to visualize health indices, anomaly indicators, and Remaining Useful Life (RUL) during vessel field trials.
- Collaborating with Amsterdam University of Applied Sciences and industry stakeholders on a funded Professional Doctorate proposal.

Computational Engineer Jan 2022 – Jan 2025
Capgemini Engineering GmbH, Munich

- Conducted 3D thermo-mechanical simulations for EUV lithography systems to evaluate structural integrity and thermal behavior under varying loads.
- Improved steady-state and transient thermal models using ANSYS, accelerating solver convergence by **20%**.
- Optimized cooling circuit designs, resulting in a **16%** increase in overall thermal management efficiency.
- Evaluated FEM results, performed sensitivity studies and hotspot analyses to maximize component lifetime.
- Applied Model Order Reduction (MOR) methodologies to decrease matrix size by **96%** while preserving 10^{-8} numerical accuracy.
- Built MATLAB and VBA automation tools to streamline encoder testing, achieving up to **40%** processing time reduction and **90%** fewer manual errors.

Master Thesis | FEM-based Strength Verification of Bolted Joints Nov 2020 – Jun 2021
Robert Bosch Automotive Steering GmbH, Schwäbisch Gmünd

- Developed an automated strength verification framework and simulation prototypes for Electronic Power Steering bolts using ABAQUS and Python, reducing analysis time by **90%**.
- Applied VDI 2230 compliant joint analysis with **less than 0.5%** deviation from physical tests through Python-based automation.
- Designed custom ABAQUS plugins and created parametric bolt models, enabling standardized and repeatable joint analysis.
- Accomplished **3**× faster design iterations by eliminating manual Excel-based workflows.
- Enhanced computational efficiency by **40%** while improving result reliability through experimental validation.

Research Assistant Sep 2018 – Oct 2020
Open Hybrid Lab Factory GmbH, Wolfsburg

- Performed advanced ANSYS simulations on FRP and hybrid materials, achieving higher accuracy in stress predictions and a **25%** thermal stress reduction through design optimization.
- Enhanced crack detection and prediction workflows using ImageJ and Python-based machine learning algorithms, with over **90%** prediction accuracy.
- Created VBA-based automation tools reducing data processing time by **50%** across research team workflows.
- Programmed **3** custom UMAT subroutines in C/C++ for ABAQUS, enabling advanced nonlinear material analysis of potting materials.

Projects

Multi-fidelity Design Optimization of Long Range BWB Aircraft Sep 2019 – Apr 2020
Technische Universität Braunschweig, Germany

- Executed multi-fidelity design optimization for a long-range Blended Wing Body aircraft, optimizing aerodynamic performance for aerospace applications.
- Built a SUAVE-based workflow integrating OpenVSP for geometry, GMSH for meshing, and SU2 for high-fidelity computational fluid dynamics.
- Implemented performance estimation for fuel efficiency and maximum takeoff weight using SUAVE's Python-based aircraft sizing algorithms.
- Reduced total aerodynamic drag by **75.95%** (from 0.018629 to 0.004480) through optimized pressure distribution and Free Form Deformation (FFD).
- Achieved a **38.96%** reduction in Maximum Takeoff Weight (MTOW) while boosting overall fuel efficiency by **35.96%**.

Education

Master of Science, M.Sc Oct 2017 – Jun 2021
Technische Universität Braunschweig, Germany
Computational Sciences in Engineering **Grade: 1.9**

Bachelor of Engineering, B.E. Aug 2012 – Aug 2016
Dayananda Sagar College of Engineering, India
Mechanical Engineering **Grade: 81%**

Languages

English - Fluent (IELTS 7.5)
German - Proficient (DSH 2/C1)
Tamil - Mother Tongue
Hindi, Kannada - Fluent

Interests

Swimming
Adventure Sports
Hiking
Motorcycling

Certifications

ANSYS APDL Programming
Deep Learning A-Z™: Hands-On ANN
Complete Modern C++ (C++11/14/17)
Machine Learning – Andrew Ng (Coursera)